

1. S&P 500

1.1. The S&P 500 is a stock market index that tracks the performance of 500 large publicly traded US companies from a wide range of sectors. The index is weighted by market capitalisation, meaning that companies with a larger total market value have a greater influence on the index. Companies included in the S&P 500 are selected by a committee based on several criteria including large market capitalisation, high liquidity, and sufficient trading volume. Firms must also be US-based and publicly listed on major U.S. exchanges. Using Python, we plot the S&P 500 index:

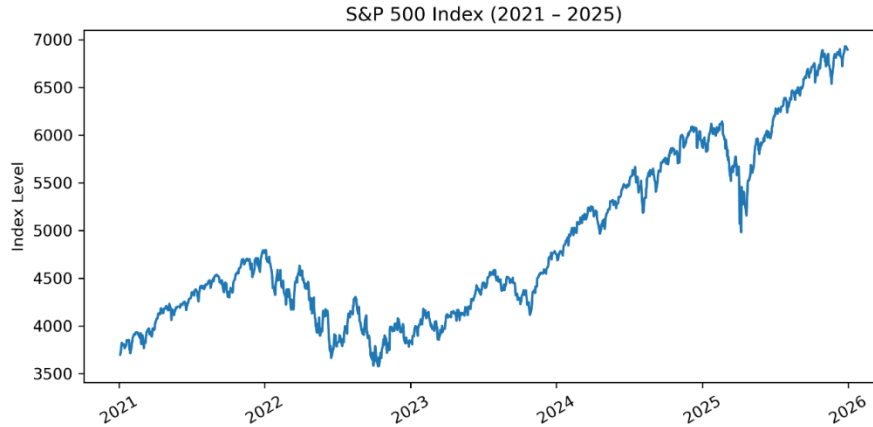


Figure 1

1.2. We assume that the S&P 500 index follows a Geometric Brownian Motion (GBM), therefore, implying that the logarithmic returns, r_t , are normally distributed:

$$r_t = (\ln S_t - \ln S_{t-1}) \sim N \left[\left(\mu - \frac{\sigma^2}{2} \right) \Delta t, \sigma^2 \Delta t \right]$$

Given observed daily price ($\Delta t = 1$) in the dataset, we compute logarithmic returns r_t then estimate the sample mean ($\bar{r}$) and variance (s^2) using Excel-based function AVERAGE() and VAR.S(), respectively. We then have: $\bar{r} = 0.0497\%$ $s^2 = 0.0114\%$

Finally, we obtain the estimate value of drift and diffusion coefficients:

$$\mu = \hat{\mu} = \bar{r} + (\sigma^2/2) = \bar{r} + (s^2/2) = 0.0497\% + (0.0114\%/2) = 0.0554\% (\text{day}^{-1})$$

$$\sigma = \hat{\sigma} = \sqrt{s^2} = \sqrt{0.0114\%} = 1.0676\% \text{ per } \sqrt{\text{day}}$$

1.3. We have the distribution of r_T , where $T = 252$ days, given by:

$$r_T = (\ln S_T - \ln S_0) \sim N [0.0554\%T, (1.0676\%)^2T] = N (13.9608\%, 2.8728\%)$$

The probability that S&P 500 will increase by 25% after one year is:

$$\begin{aligned} P(S_T \geq S_0 + 0.25S_0) &= P(S_T \geq 1.25S_0) = P(S_T/S_0 > 1.25) = 1 - P[\ln(S_T/S_0) < \ln 1.25] \\ &= 1 - \Phi \left[\frac{\ln 1.25 - \left(13.9608\% - \frac{2.8728\%}{2} \right)}{\sqrt{2.8728\%}} \right] = 1 - \Phi(0.5776) = 28.1767\% \end{aligned}$$

2. Optimal Portfolio Construction and Diversification

2.1. We have the following chart which is generated using Python:

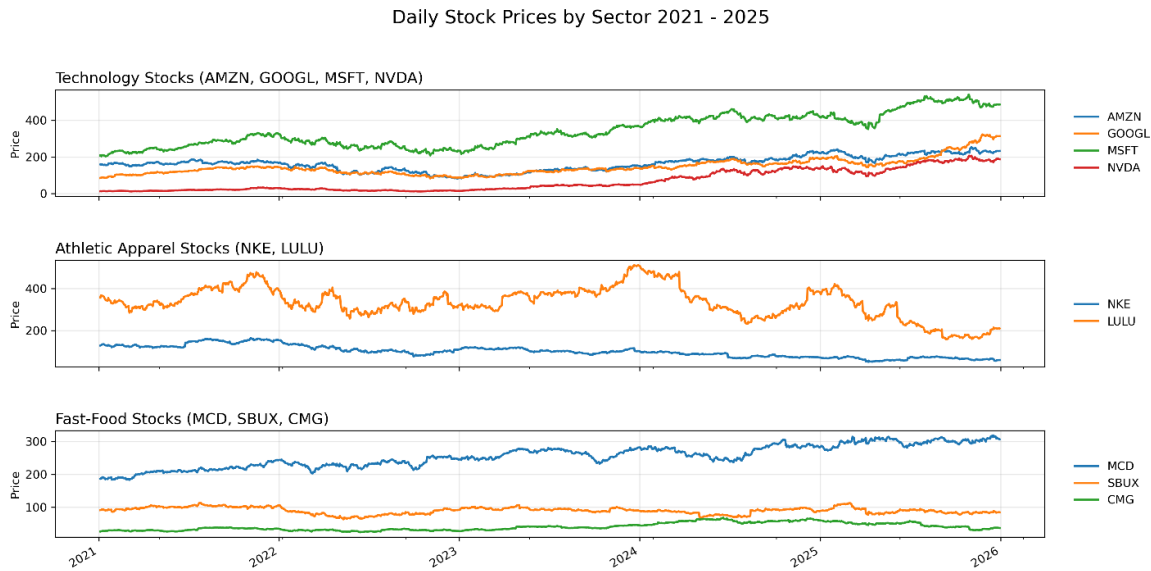


Figure 2

2.2. Since the question specifies simple returns, we compute:

$$r_{i,t} = \frac{S_{i,t} - S_{i,t-1}}{S_{i,t-1}}$$

where $S_{i,t}$ and $S_{i,t-1}$ denote the price of stock i at time t and $t-1$, respectively.

The 2nd and 3rd columns in Table 1 present the results of the sample mean and sample standard deviation of the daily returns, respectively, calculated using the Excel functions AVERAGE() and STDEV.S().

2.3. Here, the market portfolio is proxied by the S&P 500 index. We first now compute the daily return of the market at time t :

$$r_{m,t} = \frac{S_{m,t} - S_{m,t-1}}{S_{m,t-1}}$$

where $S_{m,t}$ and $S_{m,t-1}$ denote the value of market index at time t and $t-1$, respectively.

The daily expected market return ($E(r_m)$) and the variance of daily market returns ($\text{Var}(r_m)$), are computed using the Excel functions AVERAGE() and VAR.S(), respectively.:

$$E(r_m) = 0.0554\% \quad \text{Var}(r_m) = 0.0114\%$$

We calculate each stock's β by the Excel functions COVARIANCE.S() and VAR.S() with the formula:

$$\beta_i = \frac{\text{Cov}(r_i, r_m)}{\text{Var}(r_m)}$$

where r_i denotes the daily return of stock i , r_m represents the daily return of the market portfolio, and $\text{Var}(r_m)$ captures the variability of daily market returns over the sample period.

Stocks	Sample Mean ($\bar{r}$)	Sample Std. Dev. (s)	Beta (β)	CAPM Expected Return
MSFT	0.0807%	1.6203%	1.1545	0.0621%
NVDA	0.2663%	3.2911%	2.1662	0.1061%
AMZN	0.0546%	2.2122%	1.4842	0.0765%
GOOGL	0.1229%	1.9627%	1.2688	0.0671%
NKE	– 0.0361%	2.1949%	1.0980	0.1678%
LULU	– 0.0081%	2.5856%	1.2813	0.1757%
MCD	0.0455%	1.0820%	0.4130	0.0299%
CMG	0.0489%	2.0830%	1.0451	0.0574%
SBUX	0.0128%	1.9616%	0.9965	0.0552%

Table 1

The 4th column in Table 1 presents the result of the β coefficients for each stock.

The CAPM expected daily return for each stock was estimated using the CAPM formula:

$$E\left(r_i^{\text{daily}}\right) = \frac{r_f}{T} + \beta_i \left[E\left(r_m\right) - \frac{r_f}{T} \right] = 0.0119\% + 0.0435\% \beta_i$$

where r_f denotes the annual risk-free rate (3%), T represents the number of trading days in a year (252), and $E\left(r_m\right)$ is the expected daily market return, estimated to be 0.0554%.

The 5th column in Table 1 presents the result of the CAPM return daily returns for each stock. We have some comments:

- The CAPM results indicate a positive relationship between beta and expected return, where beta measures systematic risk, while standard deviation σ captures total risk. For example, NVDA has both a high beta (2.1662) and high volatility (3.2911%), suggesting that a large proportion of its total risk is driven by market movements. Using the variance decomposition (*), approximately 49% of NVDA's total risk is systematic, confirming its strong sensitivity to the market. In contrast, LULU and NKE exhibit relatively high volatility but only moderate beta values. This indicates that a significant portion of their risk is idiosyncratic (firm-specific) rather than market-related. Meanwhile, MCD has both low beta (0.4130) and low volatility (1.0820%), suggesting low exposure to systematic risk.
- When comparing CAPM expected returns with realised mean returns, several discrepancies are observed. NVDA's realised return (0.2663%) is substantially higher than its CAPM prediction (0.1061%), indicating strong outperformance relative to its level of systematic risk. Similarly, MSFT and GOOGL exhibit realised returns that are slightly above their CAPM predictions. In contrast, NKE (– 0.0361%) and LULU (– 0.0081%) have negative realised returns despite positive CAPM expected returns (0.1678% and 0.1757%, respectively), indicating underperformance. These deviations suggest that factors beyond market risk, such as firm-specific conditions or industry effects, play an important role in determining actual returns.

- Finally, while CAPM captures the relationship between systematic risk and expected return, it does not fully explain the actual returns as total risk includes both systematic and idiosyncratic components

(*): $\text{Var}(r_i) = \beta_i^2 \sigma_M^2 + \text{Var}(\epsilon_i)$ where the first term captures systematic risk driven by market movements, and the second term captures idiosyncratic risk specific to the firm.

2.4. For each pair of stocks i and j , the correlation coefficient is defined as:

$$\rho_{i,j} = \frac{\text{Cov}(r_i, r_j)}{\sigma_i \sigma_j}$$

where $\text{Cov}(r_i, r_j)$ denotes the covariance of daily returns of stocks i and j , respectively, while σ_i and σ_j are their standard deviations of their respective returns.

The correlation measures the strength and direction of the linear relationship between 2 stocks, taking values between -1 and 1 . The resulting matrix is symmetric and displays the pairwise correlations between all nine stocks where the diagonal elements are equal to 1 . Figure 3 represents the correlation matrix which is generated using the Data Analysis ToolPak (Correlation function).

	AMZN	GOOGL	LULU	MCD	NKE	CMG	SBUX	MSFT	NVDA
AMZN	100.0000%	61.9300%	44.9300%	18.0300%	40.5600%	45.2400%	35.6700%	65.4700%	55.7100%
GOOGL	61.9300%	100.0000%	35.1500%	21.1400%	35.2300%	36.9500%	31.1400%	64.4300%	52.7600%
LULU	44.9300%	35.1500%	100.0000%	26.6700%	51.4300%	39.9200%	38.0700%	41.8300%	36.7400%
MCD	18.0300%	21.1400%	26.6700%	100.0000%	27.5700%	29.2600%	35.5100%	25.2100%	11.6200%
NKE	40.5600%	35.2300%	51.4300%	27.5700%	100.0000%	33.7400%	47.5900%	35.7000%	30.2700%
CMG	45.2400%	36.9500%	39.9200%	29.2600%	33.7400%	100.0000%	35.6300%	50.4100%	41.5300%
SBUX	35.6700%	31.1400%	38.0700%	35.5100%	47.5900%	35.6300%	100.0000%	34.9400%	32.5800%
MSFT	65.4700%	64.4300%	41.8300%	25.2100%	35.7000%	50.4100%	34.9400%	100.0000%	63.0400%
NVDA	55.7100%	52.7600%	36.7400%	11.6200%	30.2700%	41.5300%	32.5800%	63.0400%	100.0000%

Figure 3

2.5. For each portfolio, the return and risk are computed as:

$$E(R_p) = \sum_{i=1}^n w_i E(r_i) \quad \sigma_p = \sqrt{w^T \Sigma w}$$

where w_i represents the weight of stock i , $E(r_i)$ is its mean daily return, and Σ denotes the covariance matrix of stock returns, constructed using the Covariance tool in Excel's Data Analysis ToolPak.

The minimum risk portfolio for each case is obtained using Excel Solver. The optimisation was set to minimise portfolio risk, with the asset weights in each case treated as decision variables. The only constraint imposed is that the weights sum to 1 , while short selling is allowed. The Solver algorithm then iteratively adjusts the weights to identify the optimal allocation that minimises portfolio variance.

The Sharpe ratio is then obtained by:

$$\text{Sharpe ratio} = \frac{E(R_p) - r_f}{\sigma_p}$$

where r_f denotes the daily risk-free rate, derived from the annual rate of 3% , assuming 252 trading days, i.e. $r_f = 0.03/252$.

Portfolio A	Weights	Portfolio B	Weights	Portfolio C	Weights	Portfolio D	Weights	Portfolio E	Weights
AMZN	6.4921%	NKE	66.4841%	CMG	9.7340%	MSFT	22.9817%	NVDA	8.0333%
GOOGL	25.4393%	LULU	33.5159%	SBUX	8.5164%	NKE	4.4284%	LULU	29.6020%
MSFT	82.2977%			MCD	81.7499%	MCD	72.5900%	CMG	62.3648%
NVDA	– 14.2290%								

Table 2

Portfolio	Sharpe Ratio	Portfolio Risk	Portfolio	Sharpe Ratio	Portfolio Risk
A	3.3542%	1.5343%	D	3.8477%	0.9898%
B	2.0439%	2.0439%	E	1.9886%	1.8890%
C	2.9868%	1.0414%	Full	3.53%	0.88%

Table 3

The minimum risk values and corresponding Sharpe ratios for each portfolio are summarised in Table 3, whereas the optimal asset weights for each portfolio are detailed in Table 2:

- Portfolio A reaches the risk of 1.5343%, with the optimisation assigning a dominant weight to MSFT, followed by GOOGL and AMZN, while taking a short position in NVDA. Despite a relatively high Sharpe ratio (3.3542%), the overall risk remains significantly higher than the full portfolio due to strong correlations between technology stocks (see Figure 3). This suggests that the performance is likely driven by strong returns rather than diversification.
- Portfolio B performed the worst, with the highest risk of 2.0439% and a negative Sharpe coefficient of – 0.0189. This is due to both stocks having negative average returns and limited diversification benefits due to having only 2 assets, which have already performed poorly.
- Portfolio C achieves a relatively low risk of 1.0414% and a Sharpe ratio of 2.9868%, with a large allocation to MCD, followed by smaller weights in CMG and SBUX. This reflects the stability of the industry, low volatility of stocks, and moderate correlations within the sector (see Figure 3). However, the benefits of diversification remain limited due to industry concentration.
- Portfolio D achieves a low risk of 0.9898%, close to the Full Portfolio (0.88%). The relatively low risk is driven by cross-sector diversification, where lower correlations between assets reduce overall portfolio variance (see Figure 3). The Sharpe ratio (3.8447%) exceeding that of the Full Portfolio (0.0353) indicates strong risk-adjusted performance.
- Portfolio E has a higher risk of 1.8890% substantially above the Full Portfolio (0.88%), reflecting the inclusion of high-return but volatile assets. Although cross-sector diversification is present and relatively low correlations between stocks, the volatility remains high. As a result, the Sharpe ratio (1.9886%) is lower than that of Portfolio D, illustrating the trade-off between return and risk.

In conclusion, diversification across sectors reduces portfolio risk more effectively than within-sector strategies. Portfolio D achieves strong risk-adjusted performance, while Portfolio B performs poorly due to limited diversification. Portfolio E highlights the return–risk trade-off. Overall, the Full Portfolio achieves the lowest risk through broad diversification across all assets.